PolicyClaw — QA & Test Document (QATD)

Version	1.0 — 2026-04-24
Status	Active — Hackathon MVP
Event	UMHackathon 2026 — Domain 2
Paired spec	`PRD.md` v2.2, `SAD.md` v1.0

Judged against ~10% of the rubric directly (Risk Anticipation 5% + Testing Strategy 5%) and influences another ~10% via Technical Feasibility (7%).

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1. Quality Goals & Metrics

Derived from PRD §9.1:

Metric	Target	Verification
PDF upload → preview rendered	≤2 s	Manual browser stopwatch
`/api/analyze` full response	≤15 s	Network tab + `cached` badge on reruns
Monte Carlo simulator slider	60 FPS, <100 ms update	Chrome DevTools Performance panel
ClawView highlight render	≤500 ms after `/v1/clawview` resolves	Network tab + visual check
Action Plan PDF generation	<2 s from button click	Manual stopwatch
Verdict consistency across 3 identical reruns	Byte-identical verdict + confidence_score	`test_verdict_consistency.py` (F7)
Health Score calibration	Overall in 30-85 band on real policies	Manual check on 3 sample PDFs (F5)

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2. Risk Matrix

Likelihood × Impact, scale 1 (low) – 5 (critical). Score = L × I.

ID	Risk	L	I	Score	Mitigation	Owner
R1	LLM API failure or rate-limit mid-demo	3	5	15	Streaming `post_glm_with_retry` with exponential-backoff retries (default 3 attempts / 120s read timeout; ClawView Annotate tightens to 2 attempts / 30s so it degrades within ~60s instead of hanging). Demo cache serves last successful payload. Recommend stage falls back to deterministic `generate_verdict`; Extract stage falls back to `_mock_policy_xray`; Score stage falls back to `_heuristic_health_score`; ClawView/FutureClaw fall back to heuristic/mock narratives.	shell
R2	Wifi drops during the live demo	2	5	10	`backend/data/demo_cache/*.json` pre-warmed on 3 sample PDFs before the demo. Re-run serves cached payloads; UI shows "Served from demo cache" badge.	shell
R3	ClawView bbox misalignment on scanned / rasterized PDFs	4	3	12	`/v1/clawview` catches exceptions and returns a handled error. UI renders an inline message instead of blanking. A side-pane risk list is the planned fallback if bbox alignment breaks on demo day.	clawview
R4	Verdict inconsistency across reruns (F7 break)	3	4	12	Demo cache makes repeated identical inputs fully deterministic (the `gpt-5-mini` reasoning model rejects custom temperature, so determinism is enforced via the cache, not sampling settings). Covered by `backend/tests/test_verdict_consistency.py`.	shell
R5	Simulator jitter on underpowered laptops	2	3	6	Monte Carlo is pure numpy, 1000 runs complete in <50 ms on modern laptops. LLM narrative regenerates only on toggle, not every tick. Chart export is off the render path.	futureclaw
R6	Text-native PDF parse failure	3	3	9	`pdf_parser.parse_pdf_chunks` returns `[]` on non-PDF bytes; `clawview_service` handles empty-chunk case by surfacing a low-confidence mock response rather than 500. Covered by `test_extraction.py`.	clawview
R7	LLM endpoint returns malformed JSON	2	4	8	`_extract_json_from_content` tolerates code-fenced responses. Schema validation raises on invalid shapes, which the `analyze_policy_*` wrappers catch and route into the heuristic fallback with a LOW confidence band so the UI can nudge the user toward a human advisor.	shell
R8	`OPENAI_API_KEY` unset on judge's machine	2	5	10	Pipeline automatically enters mock / heuristic mode. All four stages produce valid outputs; confidence bands drop to LOW; reasons indicate the fallback. Demo still works end-to-end.	shell
R9	Scanned-only PDFs (no OCR text layer)	2	3	6	PyMuPDF returns zero text; parsing yields empty chunks; Annotate falls back to a low-confidence "limited annotation" response. OCR is deferred post-hackathon.	clawview

ID	Risk	L	I	Score	Mitigation	Owner
R10	AI hallucination passing as cited fact	3	4	12	Every `Reason` requires a `Citation` with source + quote + locator. Prompt explicitly forbids invented citations. Low confidence steers the UI to "consult a human advisor." Disclaimer on every recommendation screen.	shell
R11	Secret leakage in commit	1	5	5	`.env` in `.gitignore`. `backend/.env.example` is the only tracked env file. The OpenAI key is read via `os.getenv`, never logged.	shell

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3. Testing Strategy

Pyramid

Layer	Framework	Location	Current count	Scope
Unit (backend)	`pytest`	`backend/tests/`	**19 green**	FutureClaw Monte Carlo + narrative (13), extraction sanity (2), premium simulation sanity (2), verdict consistency / F7 (2).
Integration (manual)	`curl` + browser	—	—	End-to-end against a live local stack with 3 sample PDFs.
End-to-end (manual)	Browser	—	—	Full user journey: upload → intake → analyze → ClawView → FutureClaw → VerdictCard → PDF download → BM/EN toggle.
Offline demo	Demo cache pre-warmed	`backend/data/demo_cache/`	—	Pipeline works with wifi disabled.

New unit tests added by this worktree

File	Tests	Purpose
`test_extraction.py`	2	`parse_pdf_chunks` handles valid PDFs; returns `[]` or raises on garbage bytes (never produces spurious chunks the LLM could over-interpret).
`test_simulation.py`	2	`project_premiums` produces 3 scenarios × 10-year arrays. Pessimistic cumulative > realistic > optimistic. Breakpoint year is `None` or 1..10. Names chosen to avoid collision with `test_futureclaw.py`'s coverage of `monte_carlo_affordability` and `simulate_life_events`.

File	Tests	Purpose
`test_verdict_consistency.py`	2	F7 acceptance. Heuristic `generate_verdict` is byte-deterministic. `analyze_policy_verdict` (fallback mode) produces identical verdict + confidence across 3 reruns.

Running tests

```
pytest backend/tests/ -q    # from repo root — CI invocation
```

Expect 19 tests green.

Manual QA checklist (pre-demo walk-through)

Step	Pass criterion
Upload sample PDF 1 (medical)	Profile fields auto-populate in Step 2.
Click Analyze	`POST /api/analyze` returns in ≤ 15 s. VerdictCard + HealthScoreGauge render.
ClawView follow-up	`POST /v1/clawview` fires exactly once; PDF renders with bbox highlights.
Click a red highlight	Tooltip with plain-language explanation appears.
Toggle FutureClaw to Life Event	`POST /v1/simulate/life-event` fires once; 4 scenario bars render.
Drag Affordability slider	Chart updates at 60 FPS; `POST /v1/simulate/affordability` fires on release only.
Repeat Analyze with identical input	"Served from demo cache" badge; verdict + confidence identical.
Toggle BM ↔ EN	All AI strings re-render in target language. Static labels switch via `t()`. Disclaimer and pill labels both switch.
Download Action Summary	jsPDF file opens cleanly in <2 s; contains verdict + 3 reasons + 3 actions + disclaimer + analysis_id.
HealthScoreGauge calibration	Overall scores on 3 sample policies all in 30-85 range (PRD F5).
Disable wifi, re-run Analyze	Demo cache serves the response; UI shows cached badge; ClawView panel surfaces a handled error (not a blank page).

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4. Test Fixtures & Data

Fixture	Location	Purpose
BNM medical inflation series	`backend/data/bnm_corpus/medical_inflation.json`	Historical anchor for FutureClaw Monte Carlo.
Life-event cost medians	`backend/data/bnm_corpus/life_event_costs.json`	Seeds life-event scenarios with cited Malaysian cost data.
Demo cache (gitignored in MVP; pre-warmed for demo)	`backend/data/demo_cache/*.json`	One file per (stage, canonical args) SHA-256. Protects demo against wifi drops and keeps verdicts deterministic.
Mock-mode guarantees	`ai_service._mock_*` , `_heuristic_*`	Every public `ai_service.analyze_*` path has a non-LLM fallback — the pipeline never returns 500 from an LLM problem.

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5. CI/CD Gates — live

`.github/workflows/ci.yml` is **live on main** and runs on every push and PR to `**` branches. Two parallel jobs:

Backend job (`ubuntu-latest` , Python 3.12)

Step	Behavior on failure
`pip install -r backend/requirements.txt`	Hard fail — must install.
`pip install ruff`	Hard fail — must install.
`python -c "import app.main"` (smoke import)	**Hard gate.** CI red on import error.
`ruff check backend/`	**Informational** (`continue-on-error: true`).
`pytest backend/tests/ -q`	**Hard gate when tests present** (guarded by `hashFiles('backend/tests/**/test_*.py') != ''`).

Frontend job (`ubuntu-latest` , Node 20)

Step	Behavior on failure
`npm ci --prefix frontend`	Hard fail.

Step	Behavior on failure
`npm run lint --prefix frontend`	**Informational** (`continue-on-error: true`). Present to surface debt without blocking shipping.
`npm run build --prefix frontend`	**Hard gate.** Next.js build must complete.

Rationale for the informational-lint choice. During the 24-hour build, style debt cannot block critical path changes. Lint warnings are surfaced to developers but do not block merges; once the team has bandwidth, promote lint to a hard gate post-hackathon.

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6. Manual Verification Checklist — one row per wow-factor demo moment

Demo moment	Acceptance
ClawView overlay renders on the real PDF	≥8 highlights per demo policy (F4 acceptance); color coding is meaningful (red is not random); highlights align with text.
Click a red ClawView highlight	Tooltip appears with plain-language explanation in current language; ≤80 words.
HealthScoreGauge overall score	In 30-85 band for real Malaysian policies (F5 acceptance — not all 90+).
FutureClaw Affordability chart	Updates at 60 FPS on slider drag; `POST /v1/simulate/affordability` fires on release only.
FutureClaw Life Event toggle	4 scenarios (Cancer / Heart Attack / Disability / Death); `POST /v1/simulate/life-event` fires exactly once per event switch; narratives regenerate.
ClawView fetch after Analyze	`POST /v1/clawview` fires exactly once after `POST /api/analyze` completes.
BM/EN toggle	All AI strings re-render in target language (narrative_en ↔ narrative_bm, plain_explanation_en ↔ plain_explanation_bm); static UI strings switch via `t()` .
Action Summary PDF	jsPDF generates a one-pager in <2 s (F8 acceptance); opens cleanly in Preview/Acrobat.
VerdictCard ADD RIDER pill	Renders when `needs_rider === true && verdict === "keep"` ; reason #1 surfaces the suggested rider.

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7. Known Limitations & Defer List

Item	Reason	Reopen when
Supabase (Auth, RLS, pgvector RAG, Realtime)	PRD §8.2 + §9.3 flag as ship target, not MVP-gating.	Post-hackathon, any paid plan.
Zh + Hokkien UI	PRD P5 limits MVP to BM + EN.	Beyond Malaysian-first scope.
Voice interrogation (F7)	Scaffold endpoint stays in mock mode.	Post-hackathon product work.
OCR path for scanned PDFs	Tesseract integration deferred; current MVP covers text-native PDFs.	Post-hackathon if user research justifies.
Frontend unit / E2E suite (Vitest + Playwright)	24-hour budget excludes setup. Manual QA + CI build gate carry the frontend.	Post-hackathon; CI already has a `frontend` job that can host the new steps.
`AIFeatures.tsx` style debt	Already deleted on main (commit `6cb1a3e`). Historical baggage resolved.	N/A.

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8. Submission Evidence

At submission time this document pairs with:

- `backend/tests/` — passing pytest suite (19 tests green).
- `backend/data/demo_cache/*.json` — pre-warmed cached runs for the 3 sample policies.
- `SAD.md` — architecture reference.
- `PRD.md` — product reference.
- A 4-minute demo video recorded on Hour 23 per PRD §10 schedule.
- `.github/workflows/ci.yml` CI run showing backend + frontend green.

End of QATD.